

Principles of Microeconomics — Day 1

In-Class Problem Set: Costs of Production

This problem set accompanies the Day 1 session. Numbers differ from the lecture example so they are fresh practice. Solutions (instructor copy) are collected at the end; set `\solnfalse` in the preamble to produce a student version.

Exercise 1 — Deriving the cost curves

A firm's short-run total cost function is

$$TC(Q) = Q^3 - 6Q^2 + 15Q + 18.$$

- (a) Identify fixed cost FC and variable cost $VC(Q)$.
- (b) Write AFC , AVC , ATC and MC as functions of Q .
- (c) Find the output that minimises AVC and the value of AVC there. Verify that MC equals AVC at that output.
- (d) Is the output that minimises ATC larger or smaller than the one that minimises AVC ? Explain in one sentence.

Exercise 2 — Reading costs from a table

A firm reports the following total cost at each level of output:

Q	0	1	2	3	4	5	6
TC	48	78	100	116	132	154	188

- (a) What is the firm's fixed cost?
- (b) For each output $Q = 1, \dots, 6$, compute marginal cost MC (the change in TC), average variable cost AVC and average total cost ATC .
- (c) At which output is AVC lowest? At which output is ATC lowest?
- (d) Explain in one sentence why the minimum of ATC occurs at a higher output than the minimum of AVC .